FARM MANAGEMENT SYSTEM

Project Report

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User Story

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

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1. INTRODUCTION

1.1 Project Overview

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

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2. IDEATION PHASE

2.1 Problem Statement

Persona	Need	Root Cause	Emotional Impact
Farmer	Needs to maintain farm and crop records	Records are manual, leading to incomplete or incorrect data	Frustrated
Farm Manager	Needs to monitor productivity and sales	Data is scattered, causing delayed and inaccurate reports	Stressed
Buyer	Needs to access crop availability and pricing	Information is not updated regularly due to lack of a digital system	Dissatisfied
System Admin	Needs to manage access and security	Access control is manual, resulting in unauthorized access	Concerned

2.2 Empathy Map Canvas

Target Users: Farmers, Farm Managers, Buyers, System Administrators

- **Think & Feel:** Need accurate, timely, and reliable farm data.
- **Hear:** Feedback and requirements from agricultural officers and field workers.
- **See:** Physical registers, cluttered paper files, disconnected spreadsheets.
- **Say & Do:** Enter records manually, double-check physical files for verification.
- **Pain:** Frequent data loss, high rate of manual errors, significant processing delays.

- **Gain:** Workflow automation, fast reporting, centralized access.

2.3 Brainstorming

Selected Idea: Salesforce-based Farm Management System (FMS) leveraging cloud infrastructure, automated workflows, and robust security protocols.

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3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

The user journey follows a 7-stage workflow:

1. **Login:** Secure authenticated entry into the platform.
2. **Access App:** Navigate to the customized FMS workspace.
3. **Enter Data:** Input farm, crop, farmer, and buyer records.
4. **Register Users:** System Admins configure user roles and permissions.
5. **Generate Reports:** Run real-time data compilations.
6. **Analyze Results:** Review interactive dashboard visual analytics.
7. **Decision Making:** Formulate data-driven operational strategies.

3.2 Solution Requirements

Functional Requirements

- User Authentication & Role Management
- Farm, Crop, Farmer, and Buyer Object Management
- Input Data Entry & Automated Validation Rules
- Data Relationship Mapping (Lookups / Master-Detail)
- Real-time Report Generation
- Interactive Dashboard Visualization

Non-Functional Requirements

- **Usability:** Intuitive interface requiring minimal technical training.
- **Security:** Role-based access control and encrypted cloud storage.
- **Reliability:** High data accuracy and loss prevention.
- **Performance:** Rapid query responses and report generation.
- **Availability:** Cloud-hosted platform with high uptime.
- **Scalability:** Easily extensible for future hardware and data expansion.

3.3 Data Flow Diagram

[Salesforce UI] ——> [Custom Objects] ——> [Validation Rules] ——> [Relationships] ——> [Reports] ——> [Dashboards]

3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud Architecture
Automation	Salesforce Validation Rules & Workflows
Analytics	Salesforce Reports & Dashboards

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4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

The FMS directly addresses core operational inefficiencies by replacing manual farming logs with automated data management. Cloud storage eliminates physical record loss, interactive dashboards enhance analytical decision-making, and automated real-time reports resolve information delivery delays.

4.2 Proposed Solution

A centralized, Salesforce-based Farm Management System equipped with customized data structures, field-level validation rules, structured security profiles, and analytics suites.

4.3 Solution Architecture

1. **Presentation Layer:** Salesforce Lightning UI for intuitive record management and interaction.
2. **Application Layer:** Business logic configured via Validation Rules, Field Dependencies, and Custom Relationships.

3. **Data Layer:** Schema composed of core Custom Objects (*Farm*, *Crop*, *Farmer*, *Buyer*).
4. **Analytics Layer:** Built-in Reports and Executive Dashboards for real-time visual metrics.

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Execution Steps

- Provisioned Salesforce Developer Org.
- Designed and built Custom Objects (*Farm*, *Crop*, *Farmer*, *Buyer*).
- Configured custom tabs and bundled them within the Lightning App layout.
- Formatted schema fields and established relationships (Picklists, Lookups, Master-Detail, Formulas, Field Dependencies).
- Designed security architecture including Profiles, Roles, and User Assignment.
- Configured Organization-Wide Defaults (OWD) and custom Sharing Rules.
- Built targeted Reports and visual Dashboards.
- Conducted system and user acceptance testing.

5.2 Product Backlog Modules

- **Epic 1:** Developer Setup & Foundation
- **Epic 2:** Data Modeling & Architecture
- **Epic 3:** Business Logic & Validation Rules
- **Epic 4:** Security Architecture & Access Control
- **Epic 5:** Reporting & Analytical Dashboards

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6. PROJECT DEVELOPMENT PHASE

The Farm Management System (FMS) in Salesforce structures and unifies data across core agricultural entities.

Key Milestones Achieved

- **Org Infrastructure:** Salesforce Developer Org setup and configuration.
- **Data Modeling:** Creation of key Custom Objects (`Farm__c`, `Crop__c`, `Farmer__c`, `Buyer__c`).
- **Navigation Setup:** Custom Tab configuration for all primary entities.

- **App Packaging:** Assembled the *Farm Management System* Lightning Application.
- **Schema Refinement:** Implementation of Picklists, Lookups, Master-Detail relationships, Formula fields, and Field Dependencies.
- **Security Setup:** Created tailored Profiles and Roles (e.g., *Sales Department*, *Farm Manager*).
- **User Management:** User account provisioning and license assignments.
- **Access Control:** Designed explicit Sharing Rules for appropriate data visibility.
- **Business Intelligence:** Constructed real-time Reports (e.g., *Farmers with Farms*) and aggregated Dashboards.

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7. FUNCTIONAL & PERFORMANCE TESTING

The system underwent rigorous testing cycles across key execution parameters:

- **Record Lifecycle:** Validated creation, updates, and deletion across all custom objects.
- **Data Integrity:** Verified input validation rules against edge-case data entries.
- **Relationship Integrity:** Checked parent-child cascade behaviors and formula calculations.
- **Security Rules:** Tested profile-based restricted access and role-hierarchy visibility.
- **Sharing Rules:** Verified cross-departmental data access boundaries.
- **Reporting & Dashboards:** Confirmed real-time accuracy and visual rendering of analytical outputs.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized System:** Single source of truth for all agricultural operations.
- **Paperwork Reduction:** Replaces physical logs with cloud records.
- **Improved Accuracy:** Field validations prevent erroneous data inputs.
- **Enhanced Monitoring:** Real-time visibility into crop, sales, and farming status.
- **Actionable Analytics:** Embedded reports and dashboards simplify decision-making.

Disadvantages

- **Connectivity Dependent:** Requires active internet access for real-time cloud sync.
- **Licensing:** Dependent on Salesforce ecosystem subscriptions and licenses.
- **Offline Limits:** Restricted functionality during disconnected field operations.

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9. CONCLUSION

The Farm Management System successfully digitizes agricultural operations through a robust Salesforce deployment. By centralizing data storage, enforcing validation rules, and delivering real-time analytical dashboards, the system minimizes operational errors, eliminates manual paperwork, and empowers stakeholders with data-driven decision-making tools.

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10. FUTURE SCOPE

- **Mobile App Integration:** Native mobile capabilities via Salesforce Mobile for field personnel.
- **Predictive AI Analytics:** AI-driven crop yield and harvest forecasting models.
- **Weather API Integration:** Live weather feed sync for automated irrigation and task planning.
- **IoT Sensor Integration:** Direct connectivity with soil moisture and field environmental sensors.
- **Government Schemes:** Automated tracking and application workflows for agricultural subsidies.
- **Market Price Intelligence:** Live market integration for real-time pricing and optimal selling windows.

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11. APPENDIX

- **Source Code / Configuration:** Salesforce Metadata API definitions, Validation Rules, and Custom Object Schemas.
- **Dataset Schema:** Sample test records for Farms, Crops, Farmers, and Buyers.
- **Platform & Tools:** Salesforce Developer Org, Lightning App Builder, Reports & Dashboard Engine.